

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

PHY 412 — Past Questions & Materials
by Renance

Course: PHY 412
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 2
Levels: 400

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Paper 1 of 2 — PHY 412 Analytical Mechanical II

Level 400 • First Semester • PastQuestion

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CS /608§ | 2077

Federal University of Petroleum Resources Effurun, Nigeria.

College of Science, Department of Physics.

, First Semester Examination

Academic Session: 2023/2024

Course Title: Analytical Mechanics IT

Course Code: PHY412 (3 units)

Time Allowed: 22 hrs. Date 04/05/2024

Instructions: Answer Question 1 and Any other Three Questions. Complete and ordered " answers are preferred to fragments. Note that all symbols have their usual meanings.

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QUESTION 1

a) For a system consisting of N particles moving freely in space, define the following terms:

(i) degrees of freedom, (ii) generalized coordinates, (iii) generalized velocities, and (iv) generalized momentum. (5 marks)

b) State and explain the principles of Virtual work, and D'Alembert's principle. (3 marks)

c) Two engine blocks of masses m , and m_2 are located on a frictionless double inclined plane and connected by an inextensible massless string passing over a smooth peg as shown below

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(i) Use the principle of virtual work to show that at equilibrium, $m_1 \sin \theta_1 = m_2 \sin \theta_2$.

: (6 marks)

(ii) Use D'Alembert's principle to describe the motion of the engine blocks. (8 marks)

QUESTION 2

a) Assuming the Hamiltonian $H = H(p, q, t)$, derive the Hamilton's equations of motion and * mention in detail the physical significance of the Hamiltonian, H . (8 marks)

b) For a given particle of mass m , derive the first and second sets of the Hamilton's equations together with the time-dependent equation $\dot{q}$ in Cartesian coordinates, $\dot{q}$ (8 marks)

QUESTION 3

is for writing the Lagrange's equations in problem solving.

(8 marks)

tion of a single particle of mass m , in Polar coordinates using the

(8 marks)

a) Explicitly state the eight (8) rule.

b) Find the equations of mo

Lagrange's formulations,

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QUESTION 4 7"

- a) (i) State four drawbacks of Newtonian Mechanics that led to the development of alternative methods to formulation of classical mechanics. (2 marks)
- (ii) What are the new methods to eliminate or minimize the drawbacks in Question a) above. (2 marks)
- b) It is often said that the Hamiltonian methods are more powerful than the Lagrangian methods. Briefly justify this statement. (4 marks) .
- c) Use Hamilton's principle of least action to find the equations of motion of one-dimensional Harmonic Oscillator. (8 marks)

QUESTION 5

Two equal masses (m) are connected by springs having equal spring constants as shown below, so that the masses are free to slide on a frictionless table PQ. The points P & Q to which the ends of the springs are attached are fixed.

Use Lagrange's equation to obtain the equations of motion for the vibrating masses,

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(16 marks)

QUESTION 6

- a) (i) What is a rigid body? (ii) Let a rigidly connected system of masses rotate with angular velocity ω and about an axis passing through O. Derive the inertia tensor of the system. : (11 marks)

- b) What is principal axis of inertia? (4 mark)

* o

- c) Find the (i) moments of inertia, and (ii) product of inertia of a uniform square plate of length 'a' about the x, y and z axes. (4 marks)

Paper 2 of 2 — Analytical mechanics II PHY 412

Level 400 • First Semester • PastQuestion

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Federal University of Petroleum Resources Effurun, Nigeria.
College of Science, Department of Physics.
First Semester Examination

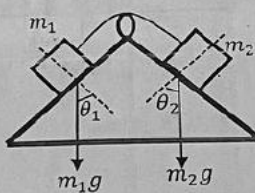
Academic Session: 2023/2024
Course Title: Analytical Mechanics II
Course Code: PHY412 (3 units)
Time Allowed: 2½ hrs.

Date 04/05/2024

Instructions: Answer Question 1 and Any other Three Questions. Complete and ordered answers are preferred to fragments. Note that all symbols have their usual meanings.

QUESTION 1

- For a system consisting of N particles moving freely in space, define the following terms: (i) degrees of freedom, (ii) generalized coordinates, (iii) generalized velocities, and (iv) generalized momentum. (5 marks)
- State and explain the principles of Virtual work, and D'Alembert's principle. (3 marks)
- Two engine blocks of masses m_1 and m_2 are located on a frictionless double inclined plane and connected by an inextensible massless string passing over a smooth peg as shown below



- Use the principle of virtual work to show that at equilibrium, $m_1 \sin \theta_1 = m_2 \sin \theta_2$. (6 marks)
- Use D'Alembert's principle to describe the motion of the engine blocks. (8 marks)

QUESTION 2

- Assuming the Hamiltonian $H = H(p_k, q_k, t)$, derive the Hamilton's equations of motion and mention in detail the physical significance of the Hamiltonian, H . (8 marks)
- For a given particle of mass m , derive the first and second sets of the Hamilton's equations together with the time-dependent equation in Cartesian coordinates. (8 marks)

QUESTION 3

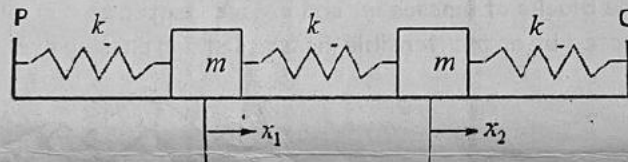
- Explicitly state the eight (8) rules for writing the Lagrange's equations in problem solving. (8 marks)
- Find the equations of motion of a single particle of mass m , in Polar coordinates using the Lagrange's formulations. (8 marks)

QUESTION 4

- a) (i) State four drawbacks of Newtonian Mechanics that led to the development of alternative methods to formulation of classical mechanics. (2 marks)
- (ii) What are the new methods to eliminate or minimize the drawbacks in Question a(i) above. (2 marks)
- b) It is often said that the Hamiltonian methods are more powerful than the Lagrangian methods. Briefly justify this statement. (4 marks)
- c) Use Hamilton's principle of least action to find the equations of motion of one-dimensional Harmonic Oscillator. (8 marks)

QUESTION 5

Two equal masses (m) are connected by springs having equal spring constants as shown below, so that the masses are free to slide on a frictionless table PQ. The points P & Q to which the ends of the springs are attached are fixed. Use Lagrange's equation to obtain the equations of motion for the vibrating masses.



(16 marks)

QUESTION 6

- a) (i) What is a rigid body? (ii) Let a rigidly connected system of masses rotate with angular velocity ω and about an axis passing through O. Derive the inertia tensor of the system. (11 marks)
- b) What is principal axis of inertia? (1 mark)
- c) Find the (i) moments of inertia, and (ii) product of inertia of a uniform square plate of length 'a' about the x , y and z axes. (4 marks)